PUFormer: A Physics-Unfolded Transformer With Exact Observation Operators for Hyperspectral and Multispectral Image Fusion

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***Abstract*—Deep unfolding networks for hyperspectral and multispectral image fusion alternate data-consistency steps with learned priors. Inside these steps, the observation operators are usually learned or implemented with the zero padding of convolution layers, whereas benchmark data are simulated with a different border rule. We examine what changes when the network's operators are exactly those of the simulation, image border included. With exact operators, the ground truth is a fixed point of every data-consistency step; with a zero-border or a learned operator, it is not. We build PUFormer, an unfolded network that uses the exact blur–decimation and spectral operators and their exact adjoints, with a transposed-attention U-Net as prior. In an equal-budget benchmark on Chikusei and Pavia Centre, in which five published networks are retrained from their authors' code under one pipeline, PUFormer is the most accurate method, 0.99 and 1.00 dB PSNR above the best retrained network. A module-by-module ablation with repeated seeds shows that the data-consistency step contributes most, whereas cross-stage memory and the type of attention change PSNR by less than 0.1 dB. At the training blur, the exact operator gains only 0.08–0.22 dB over a zero border, but it is the only operator model that can be replaced at test time: given the true blur, or one estimated from the test pair, PUFormer loses at most 0.25 dB on blur widths it was not trained on, whereas zero-border and learned operators lose 0.9–3.4 dB. Code, training logs and all result files are public.**

***Index Terms*—Boundary conditions, deep unfolding, hyperspectral image fusion, multispectral image, observation model, super-resolution.**

I. Introduction

HYPERSPECTRAL imagers record hundreds of narrow spectral bands but, for a fixed photon budget, at a coarse spatial resolution; multispectral imagers do the opposite. Hyperspectral and multispectral image fusion recovers a high-resolution hyperspectral image (HR-HSI) from one observation of each kind [1]–[3]. Classical methods formulate fusion as a regularized inverse problem, using component substitution [4], coupled matrix factorization [5], subspace regularization [6] or fast Sylvester-equation solvers [7]. Deep networks have since become the dominant approach, from convolutional reconstruction networks [8], [9] to transformers [10]–[13] and diffusion models [14]–[17].

Model-guided and unfolded networks [18]–[23] bring the observation model back into the network: each stage takes a step that reduces the mismatch between the current estimate and the two observations, and a learned module then refines the estimate. How faithfully the network reproduces the observation model has received less attention than the design of the learned module. In most of these networks the blur–decimation and spectral operators are learned convolution layers, or are implemented with the zero padding that convolution layers apply at the image border. The benchmark data, however, are simulated with a fixed blur and a different border rule. The network's operator is then not the one that produced the data, and even the ground truth fails the network's own consistency check. In simulated benchmarks this mismatch is avoidable, because the forward model is known exactly. A second difficulty concerns evaluation: reported numbers on the same dataset vary widely across papers because of differences in sensor response, blur, data split, normalization and even metric definitions [3], so improvements are only meaningful within one fully specified protocol in which all methods are run.

This paper studies the operator question in a controlled way. We use an unfolded network, PUFormer, whose data-consistency steps apply exactly the simulation operators and their exact adjoints, border included, and compare it with variants that use a zero-border or a learned operator, or no data step at all, all trained with the same budget. We then compare PUFormer with published networks that we retrain ourselves under one protocol on two airborne benchmarks. Our contributions are:

1) **The role of exact operators in unfolding.** We show that with exact operators the ground truth is a fixed point of every data-consistency step, and we quantify how far a zero-border or a learned operator moves it (Theorem 3). Experimentally, the exact operator yields a small but consistent accuracy gain, a reconstruction that agrees better with its inputs, and the ability to change the blur at test time, with a known or an estimated kernel, at almost no cost.
2) **A module-by-module ablation**, with repeated seeds for the key rows, on two datasets, that identifies which parts of an unfolded network matter: the data-consistency step and the exact operator do, whereas cross-stage memory and the type of attention hardly do, and a one-stage variant with a third of the parameters loses at most 0.2 dB.
3) **An equal-budget benchmark** in which SSR-NET,

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Code, training logs and all result files are available at https://github.com/Amarnath10i/Research-On-Hyperspectral-ImageFussion. Supplementary material (training stages, robustness, metric analyses, per-image results and complete benchmark metrics) accompanies this paper.

PSRT, DCT, MIMFormer and the unfolded DHIF-Net are retrained from their authors' code alongside PUFormer on Chikusei and Pavia Centre, with one data pipeline, one training budget and one metric implementation, complemented by a comparison with the 17 methods reported in [15] and an analysis of the metrics on which PUFormer does not rank first.

II. Related Work

A. Model-based fusion

Component substitution methods such as GSA [4] inject multispectral detail into the upsampled hyperspectral bands. Unmixing-based methods such as CNMF [5] factor both observations with shared endmembers, and subspace methods such as HySure [6] regularize the HR-HSI within a low-dimensional spectral subspace, whose dimension can be estimated with HySime [24]. FUSE [7] solves the resulting quadratic problem in closed form through a Sylvester equation. These methods use the forward model explicitly but rely on hand-crafted priors.

B. Supervised deep fusion

SSR-NET [8] and GuidedNet [9] learn a direct mapping from the two observations. Transformers model long-range spatial and spectral dependencies: PSRT [10] uses pyramid shuffle-and-reshuffle attention, MIMO-SST [11] multi-input multi-output supervision, LGCT [12] local–global collaboration, DSPNet [25] a dual spatial–spectral pyramid, SSDT [13] dilated attention, and MIMFormer [26], AMSF-Net [27], ASMNet [28] and DPFormer [29] multiscale spectral–spatial designs. Recent work also studies state-space models [30], [31], mixture-of-experts routing [32] and modality interaction [33]. General image-fusion frameworks such as U2Net [34] and OTPNet [35] have also been applied to this task.

C. Model-guided and unfolded networks

Algorithm unrolling [36]–[38] turns the iterations of an optimization method into network stages. MHF-Net [18], MoG-DCN [19] and DHIF-Net [20] unfold fusion-specific solvers, MHF-Net and DHIF-Net with operators learned as convolution layers, DUNET [21] adds transformer priors for unregistered inputs, HMPNet [22] extends unfolding to three modalities, and an information-aware transformer-based unfolding network (ITU-Net) [23], a mutual-guidance unfolding network (SMGU) [39], the perceptive spectral transformer unfolding network (PSTUN) [40], which also handles arbitrary scales, and the cross-domain-aware unfolding transformer CaFormer [41] strengthen the learned proximal steps with transformer priors. PUFormer belongs to this family, and none of its prior's components is new: passing features from one stage to the next follows memory-augmented unfolding [42], spectral-wise attention inside an unfolded network appears in DAUHST [43], and the transposed attention is that of Restormer [44]. A single unfolded network that takes the blur kernel as an input, and therefore handles kernels it was not trained on, was shown for single-image super-resolution by USRNet [45].

D. Boundary conditions

In classical deconvolution, an image border handled differently by the model and by the data causes ringing that spreads inward, and reflexive (Neumann) boundary conditions, i.e., reflect padding, are a standard remedy [46]–[48]. In iterative reconstruction, a back-projection that is not the exact adjoint of the forward model changes the point to which the iteration converges [49], [50], and in convolutional networks zero padding is itself a source of spatial bias near the border [51]. In unfolded fusion networks the operators are either learned or implemented with the zero padding of convolution layers, while benchmark data are simulated with other border rules; to our knowledge, the effect of this mismatch has not been examined. Our contribution is on this side: an unfolded network whose operators, border included, are exactly those of the simulation, and an ablation that separates the exact operator from a zero-border and a learned operator under the same budget.

E. Generative fusion

Diffusion models have been conditioned on the two observations (DDIF [14]), constrained by the observation model during sampling (CDPS [16]), restricted to a learned subspace (ARGS-Diff [17]) or combined with differential attention in a two-stage conditioning scheme [15]. Null-space formulations [52] make the data-consistency part of the reconstruction exact. The two-stage conditional diffusion model of Chen *et al.* [15] holds the best published Chikusei $\times 4$ results in our survey and is our main reference; its comparison also includes the correlation-aware network CLSNet [53] and the dual cross transformer (DCTransformer) [54], denoted DCT in their table.

III. Problem Formulation and Analysis

A. Observation model

Let $\mathbf{X} \in \mathbb{R}^{B\times H\times W}$ be the HR-HSI with B bands ($B = 128$ for Chikusei and 92 for Pavia Centre). The two observations are

$$\mathbf{Y}_{\mathrm{H}} = \mathcal{D}_\theta \mathbf{X} + \mathbf{N}_{\mathrm{H}}, \qquad \mathbf{Y}_{\mathrm{M}} = \mathcal{R}_\phi \mathbf{X} + \mathbf{N}_{\mathrm{M}}, \tag{1}$$

where $\mathcal{D}_\theta = \mathcal{S}\,\mathcal{K}_\theta\,\mathcal{P}$ applies, band by band, reflect padding $\mathcal{P}$ of width p, a valid correlation $\mathcal{K}_\theta$ with a normalized $(2p+1)\times(2p+1)$ Gaussian kernel of standard deviation θ, and decimation $\mathcal{S}$ that keeps every s-th pixel starting at index zero, so $\mathbf{Y}_{\mathrm{H}} \in \mathbb{R}^{B\times H/s\times W/s}$; $\mathcal{R}_\phi$ applies to every pixel the M spectral responses of the multispectral sensor, shifted along wavelength by ϕ nm; and $\mathbf{N}_{\mathrm{H}}$, $\mathbf{N}_{\mathrm{M}}$ are independent white Gaussian noise of standard deviation ς_{H} and ς_{M}. The benchmark protocol is the special case $\theta = \sigma_0$, $\phi = 0$ without noise, with $s = 4$, $p = 3$ and $\sigma_0 = 2$, the 7×7 blur of [15]; we write $\mathcal{D} = \mathcal{D}_{\sigma_0}$ and $\mathcal{R} = \mathcal{R}_0$ for the nominal operators and keep s and p general where the analysis does not depend on them. The spectral operator $\mathcal{R} \in \mathbb{R}^{M\times B}$ integrates the spectrum with the $M = 8$ WorldView-2 band responses [55] for Chikusei and with the $M = 4$ IKONOS band responses [56] for Pavia Centre. Because the manufacturers publish the responses only as plots together with band edges, we model each band as a flat top with logistic flanks (4 nm roll-off) that pass exactly

through the published edges, at 5% response for WorldView-2, whose edges are tabulated at that level, and at 50% for the IKONOS band limits (blue 445–516, green 506–595, red 632–698 and near infrared 757–853 nm), and normalize every band to unit sum.

Exact adjoint. Every gradient step on $\|\mathcal{D}\mathbf{X}-\mathbf{Y}_\mathrm{H}\|^2$ requires $\mathcal{D}^\top = \mathcal{P}^\top\mathcal{K}^\top\mathcal{S}^\top$. $\mathcal{S}^\top$ inserts zeros and $\mathcal{K}^\top$ is a full convolution; together they are a strided transposed convolution with an output padding of $s-1$. The adjoint of reflect padding folds the padded margins back onto the pixels they were copied from. For a padded row index $j \in [0, H+2p)$ the source row is $p-j$ for $j<p$, $j-p$ in the interior and $2H-2+p-j$ in the bottom margin, hence

$$(\mathcal{P}^\top\mathbf{g})_i = \mathbf{g}_{i+p} + [1 \le i \le p]\,\mathbf{g}_{p-i} + [H-1-p \le i \le H-2]\,\mathbf{g}_{2H-2+p-i}, \tag{2}$$

applied to rows and then to columns. With (2), $\mathcal{D}$ in the network is exactly the simulation operator, and the inner-product test $\langle\mathcal{D}\mathbf{X},\mathbf{Y}\rangle = \langle\mathbf{X},\mathcal{D}^\top\mathbf{Y}\rangle$ holds to a relative error below 10^{-13} in double precision at every scale factor. $\mathcal{R}^\top$ is the transpose of the response matrix.

B. MAP estimation and its unfolding

For given operator parameters and noise levels, and a prior $p(\mathbf{X}) \propto \exp(-\lambda\Phi(\mathbf{X}))$, the maximum a posteriori estimate is

$$\hat{\mathbf{X}} = \arg\min_{\mathbf{X}} \tfrac{1}{2\varsigma_\mathrm{H}^2}\|\mathcal{D}_\theta\mathbf{X}-\mathbf{Y}_\mathrm{H}\|^2 + \tfrac{1}{2\varsigma_\mathrm{M}^2}\|\mathcal{R}_\phi\mathbf{X}-\mathbf{Y}_\mathrm{M}\|^2 + \lambda\Phi(\mathbf{X}). \tag{3}$$

Half-quadratic splitting [45] introduces an auxiliary image $\mathbf{Z}$ coupled to $\mathbf{X}$ by a penalty $\frac{\mu_k}{2}\|\mathbf{Z}-\mathbf{X}\|^2$ and alternates, for $k = 1, \dots, K$, a data step and a prior step:

$$\mathbf{Z}_k = \arg\min_{\mathbf{Z}} \tfrac{1}{2\varsigma_\mathrm{H}^2}\|\mathcal{D}_\theta\mathbf{Z}-\mathbf{Y}_\mathrm{H}\|^2 + \tfrac{1}{2\varsigma_\mathrm{M}^2}\|\mathcal{R}_\phi\mathbf{Z}-\mathbf{Y}_\mathrm{M}\|^2 + \tfrac{\mu_k}{2}\|\mathbf{Z}-\mathbf{X}_{k-1}\|^2, \tag{4}$$

$$\mathbf{X}_k = \mathrm{prox}_{(\lambda/\mu_k)\Phi}(\mathbf{Z}_k). \tag{5}$$

With $\alpha_k = 1/(\mu_k\varsigma_\mathrm{H}^2)$ and $\beta_k = 1/(\mu_k\varsigma_\mathrm{M}^2)$, the data step (4) is the symmetric positive definite linear system

$$(\mathbf{I}+\alpha_k\mathcal{D}_\theta^\top\mathcal{D}_\theta+\beta_k\mathcal{R}_\phi^\top\mathcal{R}_\phi)\,\mathbf{Z}_k = \mathbf{X}_{k-1}+\alpha_k\mathcal{D}_\theta^\top\mathbf{Y}_\mathrm{H}+\beta_k\mathcal{R}_\phi^\top\mathbf{Y}_\mathrm{M}, \tag{6}$$

which conjugate gradients solve with the exact operators and adjoints alone, and a single gradient step from $\mathbf{X}_{k-1}$ gives its linearization

$$\mathbf{Z}_k \approx \mathbf{X}_{k-1}-\alpha_k\mathcal{D}_\theta^\top(\mathcal{D}_\theta\mathbf{X}_{k-1}-\mathbf{Y}_\mathrm{H})-\beta_k\mathcal{R}_\phi^\top(\mathcal{R}_\phi\mathbf{X}_{k-1}-\mathbf{Y}_\mathrm{M}). \tag{7}$$

An unfolded network replaces the proximal map (5) by a learned module. Three properties of this derivation shape the network of Section IV. First, α_k and β_k are not independent design parameters: their ratio $\beta_k/\alpha_k = \varsigma_\mathrm{H}^2/\varsigma_\mathrm{M}^2$ weights the two observations by their noise. Second, the proximal map depends on λ/μ_k and hence on the noise level, so a learned prior should be told it. Third, the operators enter only through $\mathcal{D}_\theta$, $\mathcal{R}_\phi$ and their adjoints in (6), so operators known or estimated at test time can be used without retraining. Theorems 1 and 3 state when the operators can be identified from the observations and what an inexact operator costs.

C. Identifiability of the operators

Theorem 1 (Commutation and identifiability): Let $\mathcal{D}_\theta$ apply the same padding, blur and decimation to every band and $\mathcal{R}_\phi$ the same matrix $\mathbf{r}_\phi \in \mathbb{R}^{M\times B}$ to every pixel. (a) $\mathcal{D}_\theta\mathcal{R}_{\phi'} = \mathcal{R}_{\phi'}\mathcal{D}_\theta$ for all θ, ϕ', with $\mathcal{D}_\theta$ acting on M or B bands. Hence, for noise-free observations generated with (θ^*, ϕ^*), the criterion

$$J(\theta,\phi) = \|\mathcal{D}_\theta\mathbf{Y}_\mathrm{M} - \mathcal{R}_\phi\mathbf{Y}_\mathrm{H}\|^2 \tag{8}$$

vanishes at (θ^*, ϕ^*). (b) Let $\mathcal{G}$ be a finite set of candidate parameters on which $\theta \mapsto \mathcal{D}_\theta$ and $\phi \mapsto \mathbf{r}_\phi$ are injective, and let no column of $\mathbf{r}_\phi$ vanish. Then, for almost every $\mathbf{X}$ (with respect to the Lebesgue measure), (θ^*, ϕ^*) is the unique zero of J on $\mathcal{G}$.

Proof: (a) On vectorized images, $\mathcal{D}_\theta = \mathbf{I}_B \otimes \bar{\mathcal{D}}_\theta$ and $\mathcal{R}_\phi = \mathbf{r}_\phi \otimes \mathbf{I}_N$, with $\bar{\mathcal{D}}_\theta$ the single-band spatial operator and $N = HW$; by the mixed-product rule, $(\mathbf{I}_M \otimes \bar{\mathcal{D}}_\theta)(\mathbf{r}_\phi \otimes \mathbf{I}_N) = \mathbf{r}_\phi \otimes \bar{\mathcal{D}}_\theta = (\mathbf{r}_\phi \otimes \mathbf{I}_{N/s^2})(\mathbf{I}_B \otimes \bar{\mathcal{D}}_\theta)$. With $\mathbf{Y}_\mathrm{M} = \mathcal{R}_{\phi^*}\mathbf{X}$ and $\mathbf{Y}_\mathrm{H} = \mathcal{D}_{\theta^*}\mathbf{X}$, $\mathcal{D}_{\theta^*}\mathbf{Y}_\mathrm{M} = \mathcal{R}_{\phi^*}\mathcal{D}_{\theta^*}\mathbf{X} = \mathcal{R}_{\phi^*}\mathbf{Y}_\mathrm{H}$. (b) By (a), $J(\theta,\phi) = \|\mathbf{E}_{\theta\phi}\mathbf{X}\|^2$ with $\mathbf{E}_{\theta\phi} = \mathbf{r}_{\phi^*}\otimes\bar{\mathcal{D}}_\theta - \mathbf{r}_\phi\otimes\bar{\mathcal{D}}_{\theta^*}$. Suppose $\mathbf{E}_{\theta\phi} = \mathbf{0}$ and apply it to $\mathbf{X} = \mathbf{e}_b \otimes \mathbf{u}$, a single band b with spatial pattern $\mathbf{u}$: $(\mathbf{r}_{\phi^*})_{:b} \otimes \bar{\mathcal{D}}_\theta\mathbf{u} = (\mathbf{r}_\phi)_{:b} \otimes \bar{\mathcal{D}}_{\theta^*}\mathbf{u}$ for all $\mathbf{u}$. If $\theta \neq \theta^*$, then $\bar{\mathcal{D}}_\theta \neq c\,\bar{\mathcal{D}}_{\theta^*}$ for every scalar c: both kernels sum to one and reflect padding preserves constants, so both operators map a constant image to the same constant, which forces $c = 1$ and contradicts injectivity. Two linear maps of rank greater than one that are not scalar multiples of each other map some $\mathbf{u}$ to linearly independent images, and for this $\mathbf{u}$ the identity above forces $(\mathbf{r}_{\phi^*})_{:b} = (\mathbf{r}_\phi)_{:b} = \mathbf{0}$, a contradiction. If $\theta = \theta^*$, the identity gives $(\mathbf{r}_{\phi^*})_{:b} = (\mathbf{r}_\phi)_{:b}$ for every b, so $\phi = \phi^*$. Hence $\mathbf{E}_{\theta\phi} \neq \mathbf{0}$ for every $(\theta,\phi) \neq (\theta^*,\phi^*)$ in $\mathcal{G}$; its null space is a proper subspace and has measure zero, and so does the finite union of these null spaces. For every $\mathbf{X}$ outside it, $J(\theta,\phi) > 0$ whenever $(\theta,\phi) \neq (\theta^*,\phi^*)$. ∎

We therefore estimate (θ,ϕ) for each test pair by minimizing (8) on a grid (steps of $0.025\,\sigma_0$ and 1 nm), using only the two observations; HySure [6] uses the same relation to estimate the responses of a model-based method. With noise, $\mathbb{E}\,J(\theta,\phi) = \|\mathbf{E}_{\theta\phi}\mathbf{X}\|^2 + \frac{N}{s^2}\big(M\varsigma_\mathrm{M}^2\|\mathbf{k}_\theta\|^2 + \varsigma_\mathrm{H}^2\|\mathbf{r}_\phi\|_F^2\big)$ up to border terms, where $\mathbf{k}_\theta$ is the blur kernel. The noise term decreases with θ and barely depends on ϕ, so noise biases $\hat{\theta}$ upward by an amount that vanishes as the SNR grows. Theorem 1 also requires the family to contain the true operator; Section **??** tests anisotropic and defocus blurs and misregistered observations, which it does not.

D. Separable structure: a closed-form data step and the fusion null space

A Gaussian kernel, isotropic or aligned with the image axes, is separable, and reflect and zero padding treat rows and columns independently; decimation does too. The blur–decimation operator therefore factors exactly, border included: for the cube $\mathbf{X} \in \mathbb{R}^{B\times H\times W}$,

$$\mathcal{D}_\theta\mathbf{X} = \mathbf{X} \times_2 \mathbf{D}_\mathrm{r} \times_3 \mathbf{D}_\mathrm{c}, \qquad \mathcal{R}_\phi\mathbf{X} = \mathbf{X} \times_1 \mathbf{r}_\phi, \tag{9}$$

where $\times_n$ is the mode-n product and $\mathbf{D}_\mathrm{r} \in \mathbb{R}^{h\times H}$ and $\mathbf{D}_\mathrm{c} \in \mathbb{R}^{w\times W}$ pad, blur and decimate a single axis. Separable

operators of this form are used by tensor-factorization fusion methods [57] to identify low-rank models, under periodic borders; here they yield an exact, closed-form data step and an explicit description of what fusion cannot observe.

Theorem 2 (Diagonalization and null space): Let $\mathbf{D}_\mathrm{r}^\top\mathbf{D}_\mathrm{r} = \mathbf{U}_\mathrm{r}\mathrm{diag}(\boldsymbol{\lambda}^\mathrm{r})\mathbf{U}_\mathrm{r}^\top$, $\mathbf{D}_\mathrm{c}^\top\mathbf{D}_\mathrm{c} = \mathbf{U}_\mathrm{c}\mathrm{diag}(\boldsymbol{\lambda}^\mathrm{c})\mathbf{U}_\mathrm{c}^\top$ and $\mathbf{r}_\phi^\top\mathbf{r}_\phi = \mathbf{V}\mathrm{diag}(\boldsymbol{\nu})\mathbf{V}^\top$ be eigendecompositions, and $\mathcal{Q}(\cdot) = (\cdot)\times_1\mathbf{V}\times_2\mathbf{U}_\mathrm{r}\times_3\mathbf{U}_\mathrm{c}$. (a) $\mathcal{D}_\theta^\top\mathcal{D}_\theta$ and $\mathcal{R}_\phi^\top\mathcal{R}_\phi$ are diagonal in the orthonormal basis $\mathcal{Q}$, with eigenvalues $\lambda_i^\mathrm{r}\lambda_j^\mathrm{c}$ for every band and ν_b for every pixel. (b) The exact data step (6) is

$$\mathbf{Z}_k = \mathcal{Q}\Big(\mathcal{Q}^{-1}(\mathbf{C}_k) \oslash \big[1 + \alpha_k\lambda_i^\mathrm{r}\lambda_j^\mathrm{c} + \beta_k\nu_b\big]_{b,i,j}\Big), \tag{10}$$

with $\mathbf{C}_k$ the right-hand side of (6) and $\oslash$ the element-wise division: three mode products in each direction, $O(BHW(B+H+W))$ operations, no iteration and no periodic-border assumption. (c) The null space of the joint operator, $\mathcal{N}(\mathcal{D}_\theta)\cap\mathcal{N}(\mathcal{R}_\phi)$, is spanned by the basis tensors with $\nu_b = 0$ and $\lambda_i^\mathrm{r}\lambda_j^\mathrm{c} = 0$; its orthogonal projector is $(\mathbf{I}-\mathbf{P}_\mathrm{s})\otimes(\mathbf{I}-\mathbf{P}_\mathrm{r}\otimes\mathbf{P}_\mathrm{c})$ in mode order, with $\mathbf{P}_\mathrm{s}$, $\mathbf{P}_\mathrm{r}$, $\mathbf{P}_\mathrm{c}$ the projectors onto the row spaces of $\mathbf{r}_\phi$, $\mathbf{D}_\mathrm{r}$, $\mathbf{D}_\mathrm{c}$. If these have full row rank, its dimension is $(B-M)(HW-hw)$, a fraction $(1-M/B)(1-1/s^2)$ of all unknowns.

Proof: (a) By (9), $\mathcal{D}_\theta^\top\mathcal{D}_\theta$ applies $\mathbf{D}_\mathrm{r}^\top\mathbf{D}_\mathrm{r}$ and $\mathbf{D}_\mathrm{c}^\top\mathbf{D}_\mathrm{c}$ to the two spatial modes and the identity to the spectral mode, and $\mathcal{R}_\phi^\top\mathcal{R}_\phi$ applies $\mathbf{r}_\phi^\top\mathbf{r}_\phi$ to the spectral mode alone; each factor is diagonalized by the corresponding factor of $\mathcal{Q}$, and an identity factor by any orthonormal basis. (b) In this basis the matrix of (6) is diagonal with entries $1+\alpha_k\lambda_i^\mathrm{r}\lambda_j^\mathrm{c}+\beta_k\nu_b > 0$. (c) $\mathcal{N}(\mathcal{D}_\theta)\cap\mathcal{N}(\mathcal{R}_\phi) = \mathcal{N}(\mathcal{D}_\theta^\top\mathcal{D}_\theta + \mathcal{R}_\phi^\top\mathcal{R}_\phi)$, whose eigenvalues $\lambda_i^\mathrm{r}\lambda_j^\mathrm{c}+\nu_b$ are sums of nonnegative terms and vanish only when both terms do. The basis tensors with $\lambda_i^\mathrm{r}\lambda_j^\mathrm{c} > 0$ span the row space of $\mathbf{D}_\mathrm{r}\otimes\mathbf{D}_\mathrm{c}$, of dimension hw, and those with $\nu_b > 0$ the row space of $\mathbf{r}_\phi$, of dimension M, which gives the projector and the dimension. ∎

The closest closed form, that of FUSE [7], solves the same system by Fourier transforms under periodic borders; (10) holds for the border rule of the simulation, which Theorem 3 shows to matter, and is a differentiable layer whose operators may change from one image to the next, at the cost of eigendecompositions of an $H\times H$, a $W\times W$ and a $B\times B$ matrix. Part (c) states what any fusion method has to infer from a prior: the high spatial frequencies of the spectral directions that the multispectral sensor does not see, 88% of the unknowns for Chikusei and 90% for Pavia Centre at $\times 4$; every other component is determined by the observations. Section VI-C uses this to split the error of every method into a part that contradicts the observations and a part in this null space.

E. The cost of an inexact operator

Stack the two data terms as $\mathbf{A} = [\mathcal{D}_\theta;\mathcal{R}_\phi]$, $\mathbf{Y} = [\mathbf{Y}_\mathrm{H};\mathbf{Y}_\mathrm{M}]$ and $\mathbf{W}_k = \mathrm{diag}(\alpha_k\mathbf{I},\beta_k\mathbf{I})$, and let $\tilde{\mathbf{A}}$ be the operator a network uses in place of $\mathbf{A}$: one with a zero border, a learned one, or one with wrong parameters. Write $T(\mathbf{V}) = (\mathbf{I}+\tilde{\mathbf{A}}^\top\mathbf{W}_k\tilde{\mathbf{A}})^{-1}(\mathbf{V}+\tilde{\mathbf{A}}^\top\mathbf{W}_k\mathbf{Y})$ for the exact data step (6) applied to an estimate $\mathbf{V}$, and $G(\mathbf{V}) = \mathbf{V}-\tilde{\mathbf{A}}^\top\mathbf{W}_k(\tilde{\mathbf{A}}\mathbf{V}-\mathbf{Y})$ for its linearization (7).

Theorem 3 (Fixed point and operator mismatch): Let $\mathbf{Y} = \mathbf{A}\mathbf{X}^*$. (a) If $\tilde{\mathbf{A}} = \mathbf{A}$, then $T(\mathbf{X}^*) = G(\mathbf{X}^*) = \mathbf{X}^*$ for all $\alpha_k,\beta_k > 0$. (b) For every $\tilde{\mathbf{A}}$,

$$\|T(\mathbf{X}^*)-\mathbf{X}^*\| \le \tfrac{1}{2}\,\|\mathbf{W}_k^{1/2}(\mathbf{A}-\tilde{\mathbf{A}})\mathbf{X}^*\|, \tag{11}$$

$$\|G(\mathbf{X}^*)-\mathbf{X}^*\| \le \|\mathbf{W}_k^{1/2}\tilde{\mathbf{A}}\|\,\|\mathbf{W}_k^{1/2}(\mathbf{A}-\tilde{\mathbf{A}})\mathbf{X}^*\|. \tag{12}$$

(c) Let stage k of a network be $\mathbf{X}_k = (\mathcal{I}+\mathcal{P}_k)(T_k(\mathbf{X}_{k-1}))$, where the exact data step T_k uses $\mathbf{W}_k$ and the residual prior satisfies $\mathcal{P}_k(\mathbf{X}^*) = \mathbf{0}$ and makes $\mathcal{I}+\mathcal{P}_k$ L_k-Lipschitz. Started at $\mathbf{X}_0 = \mathbf{X}^*$, the network returns

$$\|\mathbf{X}_K-\mathbf{X}^*\| \le \tfrac{1}{2}\sum_{k=1}^{K}\Big(\prod_{j=k}^{K}L_j\Big)\|\mathbf{W}_k^{1/2}(\mathbf{A}-\tilde{\mathbf{A}})\mathbf{X}^*\|. \tag{13}$$

Proof: (a) $\tilde{\mathbf{A}}\mathbf{X}^*-\mathbf{Y} = \mathbf{0}$ makes both updates vanish. (b) With $\mathbf{B} = \mathbf{W}_k^{1/2}\tilde{\mathbf{A}}$, a direct computation gives $T(\mathbf{X}^*)-\mathbf{X}^* = (\mathbf{I}+\mathbf{B}^\top\mathbf{B})^{-1}\mathbf{B}^\top\mathbf{W}_k^{1/2}(\mathbf{A}-\tilde{\mathbf{A}})\mathbf{X}^*$; in the singular value decomposition of $\mathbf{B}$, $(\mathbf{I}+\mathbf{B}^\top\mathbf{B})^{-1}\mathbf{B}^\top$ has singular values $b_i/(1+b_i^2) \le \frac{1}{2}$. Similarly $G(\mathbf{X}^*)-\mathbf{X}^* = \mathbf{B}^\top\mathbf{W}_k^{1/2}(\mathbf{A}-\tilde{\mathbf{A}})\mathbf{X}^*$, whose norm is at most $\|\mathbf{B}\|$ times the last factor. (c) T_k is nonexpansive, since $\|T_k(\mathbf{V})-T_k(\mathbf{V}')\| = \|(\mathbf{I}+\mathbf{B}^\top\mathbf{B})^{-1}(\mathbf{V}-\mathbf{V}')\| \le \|\mathbf{V}-\mathbf{V}'\|$. With $e_k = \|\mathbf{X}_k-\mathbf{X}^*\|$ and $(\mathcal{I}+\mathcal{P}_k)(\mathbf{X}^*) = \mathbf{X}^*$, $e_k \le L_k\|T_k(\mathbf{X}_{k-1})-\mathbf{X}^*\| \le L_k\big(\|T_k(\mathbf{X}_{k-1})-T_k(\mathbf{X}^*)\|+\|T_k(\mathbf{X}^*)-\mathbf{X}^*\|\big) \le L_k(e_{k-1}+b_k)$, where b_k is the bound (11); unrolling from $e_0 = 0$ gives (13). ∎

Theorem 3 separates the two roles of the operator. With the exact operator, the ground truth is a fixed point of every data step, so the priors only have to remove the reconstruction error. With an inexact one, every data step moves the ground truth, and the priors must also undo this displacement; because it depends on the image, it cannot be removed by a fixed correction. The exact proximal step bounds the displacement by $\frac{1}{2}\|\mathbf{W}_k^{1/2}(\mathbf{A}-\tilde{\mathbf{A}})\mathbf{X}^*\|$ whatever the learned weights, whereas for the gradient step the bound grows with $\|\mathbf{W}_k^{1/2}\tilde{\mathbf{A}}\|$, which training may make large, and the step is not nonexpansive, so (13) has no counterpart for it. The theorem also explains why an exact operator can be replaced at test time: for data generated by $\mathbf{A}'$, giving the network $\tilde{\mathbf{A}} = \mathbf{A}'$ restores (a), which a learned operator cannot do.

Corollary 1 (Border rule): If $\tilde{\mathbf{A}}$ differs from $\mathbf{A}$ only in the border rule of $\mathcal{D}_\theta$, e.g., zero instead of reflect padding, $(\mathbf{A}-\tilde{\mathbf{A}})\mathbf{X}^*$ is nonzero only on the LR pixels whose $(2p+1)\times(2p+1)$ kernel support reaches the padding, i.e., within $\lceil p/s\rceil$ pixels of the LR border.

With decimation from index zero and $p < s$, only the first LR row and column are affected, a fraction $(h+w-1)/(hw)$ of an $h\times w$ LR image: 3% of a 64×64 image but 12% of the 16×16 LR patches on which networks are trained. The displacement (11) is then small, which matches the small accuracy gap at the training blur (Section VI-A), whereas a learned operator differs from $\mathbf{A}$ everywhere. In real images the true border rule is unknown [58], and Corollary 1 bounds what any fixed choice can cost; the main role of exactness is the one stated by Theorem 3: an operator that can be supplied, estimated and replaced. Section **??** measures the displacement and the test PSNR as functions of the operator residual $\|(\tilde{\mathbf{A}}-\mathbf{A})\mathbf{X}^*\|$.

IV. PUFormer

A. Unfolded iterations

PUFormer (Fig. 1) is the HQS iteration of Section III-B unrolled for $K = 3$ stages, starting from the bicubic upsampling [59] $\mathbf{X}_0$ of $\mathbf{Y}_\mathrm{H}$. Stage k (Fig. 1, bottom) first performs the data step (4) with the exact operators, either linearized,

$$\begin{aligned}\mathbf{Z}_k = \mathbf{X}_{k-1} &- s^2\alpha_k\,\mathcal{D}_\theta^\top(\mathcal{D}_\theta\mathbf{X}_{k-1} - \mathbf{Y}_\mathrm{H}) \\ &- \beta_k\,\mathcal{R}_\phi^\top(\mathcal{R}_\phi\mathbf{X}_{k-1} - \mathbf{Y}_\mathrm{M}),\end{aligned} \tag{14}$$

or exactly, in closed form by (10) (Fig. 2; by conjugate gradients on (6) if the kernel is not separable). The weights $\alpha_k = \mathrm{softplus}(a_k)$ and $\beta_k = \mathrm{softplus}(b_k)$ stand for $1/(\mu_k\varsigma_\mathrm{H}^2)$ and $1/(\mu_k\varsigma_\mathrm{M}^2)$ in (6) and are learned per stage; the factor s^2 compensates the $1/s^2$ gain of $\mathcal{D}^\top\mathcal{D}$ on smooth content under $\times s$ decimation, so that both terms start at a comparable scale. The prior step (5) is a learned residual,

$$\begin{aligned}(\Delta_k, \mathbf{F}_k) &= \mathcal{P}_k\big([\mathbf{Z}_k, \mathbf{Y}_\mathrm{M}, s^2\mathcal{D}^\top(\mathcal{D}\mathbf{Z}_k - \mathbf{Y}_\mathrm{H})],\, \mathbf{F}_{k-1}\big), \\ \mathbf{X}_k &= \mathbf{Z}_k + \Delta_k,\end{aligned} \tag{15}$$

where the third input, the gradient of the LR-HSI data term at $\mathbf{Z}_k$, tells the prior where the updated estimate still disagrees with $\mathbf{Y}_\mathrm{H}$, and $\mathbf{F}_{k-1}$ are the final decoder features of the previous stage (cross-stage memory, merged by a 1×1 convolution). A variant also gives the prior the back-projected HR-MSI residual $(B/M)\,\mathcal{R}_\phi^\top(\mathcal{R}_\phi\mathbf{Z}_k - \mathbf{Y}_\mathrm{M})$. The last convolution of every prior is initialized to zero, so training starts from the pure HQS iteration with $\mathcal{P}_k = \mathbf{0}$, which leaves the ground truth unchanged as required by Theorem 3(c). The image path, including all operator applications, stays in 32-bit floating point; only the priors run in 16-bit mixed precision [60].

Learned spectral data step. In the basis of Theorem 2, the exact step multiplies every component of the data residual by the gain $1/(1+\alpha_k\lambda_i^\mathrm{r}\lambda_j^\mathrm{c}+\beta_k\nu_b)$, the filter of a white Gaussian prior. Natural images are not white, and noise does not harm all components equally, so we let two small networks modulate these gains:

$$\begin{aligned}\mathbf{Z}_k &= \mathbf{X}_{k-1} + \mathcal{Q}\big(\mathcal{Q}^{-1}(\mathbf{G}_k)\odot\mathbf{\Gamma}_k\big), \\ \Gamma_{k,bij} &= \frac{\exp\big(\ln 2\,\tanh(u_k(\lambda_i^\mathrm{r}\lambda_j^\mathrm{c},\mathbf{c}) + v_k(\nu_b,\mathbf{c}))\big)}{1+\alpha_k\lambda_i^\mathrm{r}\lambda_j^\mathrm{c}+\beta_k\nu_b},\end{aligned} \tag{16}$$

where $\mathbf{G}_k = \alpha_k\mathcal{D}_\theta^\top(\mathbf{Y}_\mathrm{H} - \mathcal{D}_\theta\mathbf{X}_{k-1}) + \beta_k\mathcal{R}_\phi^\top(\mathbf{Y}_\mathrm{M} - \mathcal{R}_\phi\mathbf{X}_{k-1})$ is the data residual and u_k, v_k are two-layer perceptrons of the log-eigenvalue and of the observation code $\mathbf{c}$ introduced below, initialized to zero so that training starts from the exact step (10). The gain is separable, so a stage evaluates the networks $HW + B$ times rather than BHW times.

Proposition 1: For all u_k, v_k: (a) with exact operators and noise-free data, $\mathbf{X}^*$ is a fixed point of (16); (b) (16) is nonexpansive; (c) with an inexact operator $\tilde{\mathbf{A}}$, it moves $\mathbf{X}^*$ by at most $\|\mathbf{W}_k^{1/2}(\mathbf{A}-\tilde{\mathbf{A}})\mathbf{X}^*\|$, twice the bound (11) and independent of the weights.

Proof: Let $\varsigma_{bij} = \alpha_k\lambda_i^\mathrm{r}\lambda_j^\mathrm{c} + \beta_k\nu_b$, the eigenvalues of $\tilde{\mathbf{A}}^\top\mathbf{W}_k\tilde{\mathbf{A}}$, and e^f the numerator of Γ, so $\frac{1}{2} < e^f < 2$. (a) $\mathbf{G}_k = \mathbf{0}$ at $\mathbf{X}^*$. (b) The step maps a difference $\mathbf{V} - \mathbf{V}'$ to $\mathcal{Q}\big((1-\Gamma\varsigma)\odot\mathcal{Q}^{-1}(\mathbf{V}-\mathbf{V}')\big)$, and $0 \le \Gamma\varsigma = e^f\varsigma/(1+\varsigma) < 2$, so every factor lies in $(-1, 1]$. (c) The displacement is $\mathcal{Q}\big(\Gamma\odot\mathcal{Q}^{-1}(\mathbf{B}^\top\mathbf{e})\big)$ with $\mathbf{B} = \mathbf{W}_k^{1/2}\tilde{\mathbf{A}}$ and $\mathbf{e} = \mathbf{W}_k^{1/2}(\mathbf{A}-\tilde{\mathbf{A}})\mathbf{X}^*$; its operator norm is $\max e^f\sqrt{\varsigma}/(1+\varsigma) \le 2\cdot\frac{1}{2} = 1$. ∎

Theorem 3(a) and (c) therefore hold for the learned step too: it can reweight how much the network trusts each observed component, but it cannot move a correct estimate or amplify errors.

Algorithm 1 PUFormer inference (unrolled HQS)

Require: $\mathbf{Y}_\mathrm{H}$, $\mathbf{Y}_\mathrm{M}$; operator parameters (θ, ϕ), given or $\arg\min_\mathcal{G} J$ of (8)

build the exact $\mathcal{D}_\theta, \mathcal{D}_\theta^\top, \mathcal{R}_\phi, \mathcal{R}_\phi^\top$ (reflect border, (2))
$\mathbf{X}_0 \leftarrow \mathrm{bicubic}_{\times s}(\mathbf{Y}_\mathrm{H})$, $\mathbf{F}_0 \leftarrow \varnothing$
for $k = 1, \dots, K$ **do**
  $\mathbf{Z}_k \leftarrow$ data step: (14), or CG on (6) ▷ fp32
  $\mathbf{r}_k \leftarrow s^2\,\mathcal{D}^\top(\mathcal{D}\mathbf{Z}_k - \mathbf{Y}_\mathrm{H})$
  $(\Delta_k, \mathbf{F}_k) \leftarrow \mathcal{P}_k([\mathbf{Z}_k, \mathbf{Y}_\mathrm{M}, \mathbf{r}_k], \mathbf{F}_{k-1})$ ▷ fp16
  $\mathbf{X}_k \leftarrow \mathbf{Z}_k + \Delta_k$
end for
return $\mathbf{X}_K$

B. Transposed-attention prior

Each prior (Fig. 3) is a three-level U-Net [61] with widths 48, 96 and 192, two, three and four transformer blocks on the encoder levels and bottleneck, mirrored decoder levels, and two refinement blocks. A block (Fig. 4(a)) follows Restormer [44]: multi-Dconv-head transposed attention (Fig. 5) computes queries, keys and values with 1×1 and depth-wise 3×3 convolutions, normalizes each feature map over space, and forms a $c' \times c'$ attention matrix across channels per head (1, 2 and 4 heads on the three levels) with a learned temperature. Attention is thus computed between spectral feature maps rather than between spatial positions, its cost is linear in the number of pixels, and the receptive field of every block covers the whole image. A gated depth-wise feed-forward network with expansion 2.66 follows (Fig. 4(b)). The three priors hold 3.37–3.38 M parameters each; the whole network has 10.13 M parameters for Chikusei and 9.99 M for Pavia Centre, whose input and output convolutions are smaller.

C. Training objective

With ground truth $\mathbf{X}$, the loss is

$$\begin{aligned}\mathcal{L} = \|\mathbf{X}_K - \mathbf{X}\|_1 &+ 0.1\sum_{k<K}\|\mathbf{X}_k - \mathbf{X}\|_1 \\ &+ \lambda_\mathrm{SAM}\,\mathrm{SAM}(\mathbf{X}_K, \mathbf{X}) + \lambda_\mathrm{S}\big(1 - \mathrm{SSIM}(\mathbf{X}_K, \mathbf{X})\big),\end{aligned} \tag{17}$$

with mean absolute errors, deep supervision [62] of the intermediate stages, the mean spectral angle in radians, and a band-wise Gaussian SSIM [63] whose data range is the maximum of each training patch. We use $\lambda_\mathrm{SAM} = 0.05$ and $\lambda_\mathrm{S} = 0.1$; both terms are computed in 32-bit precision. An exponential moving average of the weights (decay 0.999) [64] is used for validation and testing.

D. Training for a family of observation models

Trained with one fixed operator pair on noise-free data, the priors adapt to one blur, one spectral response and no noise,

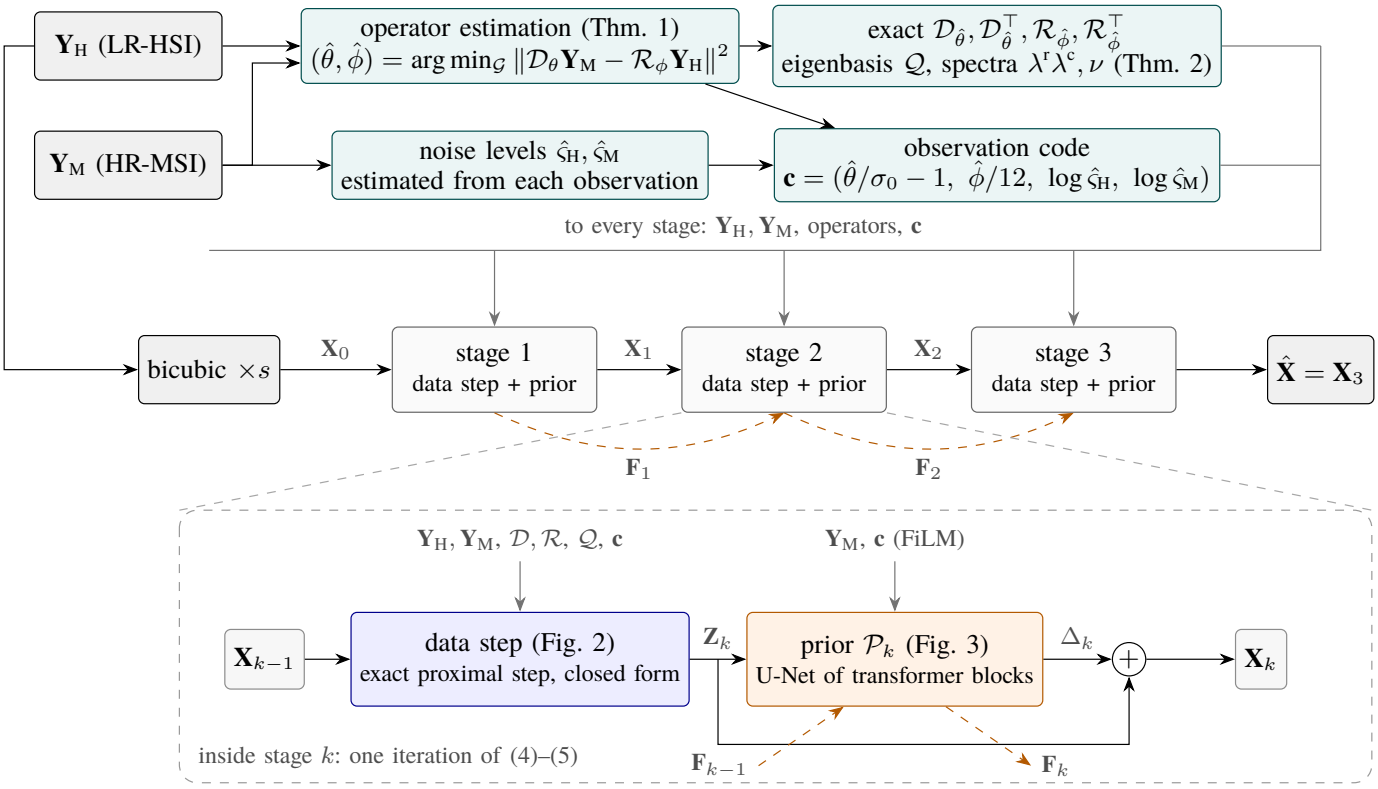


Fig. 1. PUFormer: the HQS iteration of Section III-B unrolled for three stages. Top: the observation model is set up once per image, from given parameters or from the estimate of Theorem 1, together with the noise levels and the observation code $\mathbf{c}$. Middle: every stage receives the previous estimate, both observations, the operators and $\mathbf{c}$, and passes its decoder features to the next stage (memory, dashed). Bottom: inside a stage, an exact data step on both observations is followed by a learned residual prior.

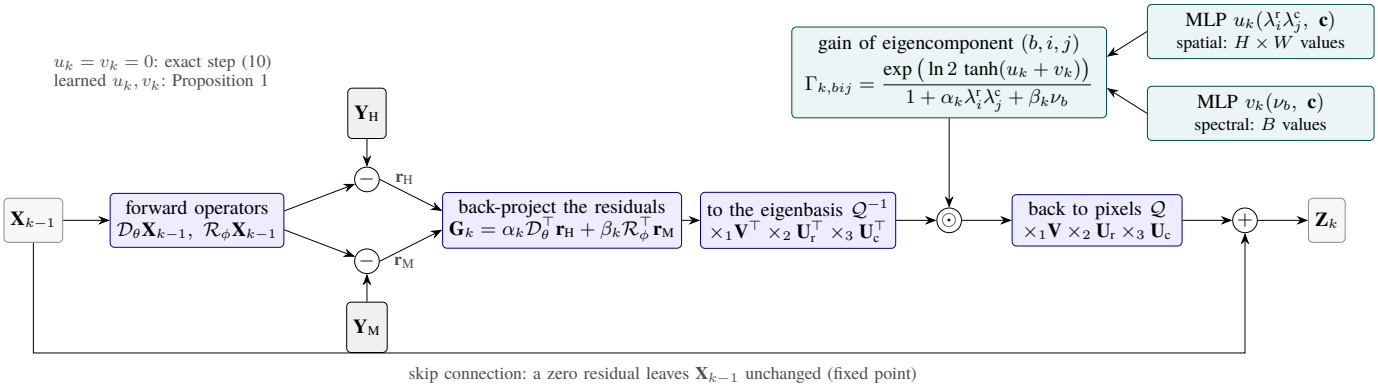


Fig. 2. The data step of stage k. The residuals of both observations are back-projected with the exact adjoints, transformed to the eigenbasis of the separable operators by one product per tensor mode (spectral, rows, columns), scaled per eigencomponent and transformed back. The gains depend on the spatial eigenvalues $\lambda_i^\mathrm{r}\lambda_j^\mathrm{c}$ of $\mathcal{D}_\theta^\top \mathcal{D}_\theta$, the spectral eigenvalues ν_b of $\mathcal{R}_\phi^\top \mathcal{R}_\phi$, the weights α_k, β_k and, through two small networks, the observation code $\mathbf{c}$.

and Section VI-F shows the consequences: a shifted spectral response or noise is not compensated by giving the network the true operator. Because the operators enter PUFormer only through its data steps, the same network can instead be trained for a family of observation models. For every batch we draw a blur width $\sigma \sim \mathcal{U}[0.5, 2]\,\sigma_0$, a shift $\delta \sim \mathcal{U}[-12, 12]$ nm of the spectral responses and, with probability 0.5, Gaussian noise on both observations at an SNR drawn from $\mathcal{U}[30, 50]$ dB; the batch is simulated with $(\mathcal{D}_\sigma, \mathcal{R}_\delta)$, and the same operators are used in the data steps. At test time the operators are given, or estimated from the test pair by minimizing (8). Training on randomized degradations is a standard remedy in blind super-resolution [65], [66], and USRNet [45] passes the blur kernel to the data step of an unfolded single-image super-resolution network. We extend this to the two operators of fusion, with exact borders and with both operators estimated from the observation pair, and compare it with randomized training alone, in which the data steps keep the nominal operator.

Observation-model conditioning. The data steps then know the operator, but the prior does not, and neither knows the noise level, although the HQS derivation (Section III-B) shows that the weights α_k, β_k should follow the noise of

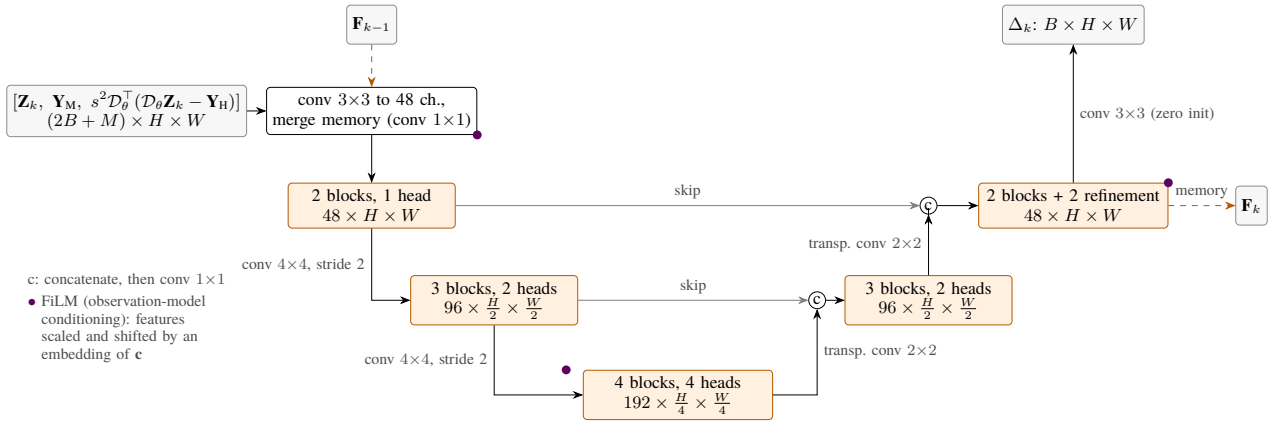


Fig. 3. The prior $\mathcal{P}_k$ of stage k (widths for Chikusei, $B=128$, $M=8$): a three-level U-Net of transformer blocks (Fig. 4). The encoder halves the resolution twice with strided convolutions, the decoder restores it with transposed convolutions and merges the encoder features through skip connections; the previous stage's decoder features enter after the input convolution, and the last convolution starts at zero so that training starts from the HQS iteration.

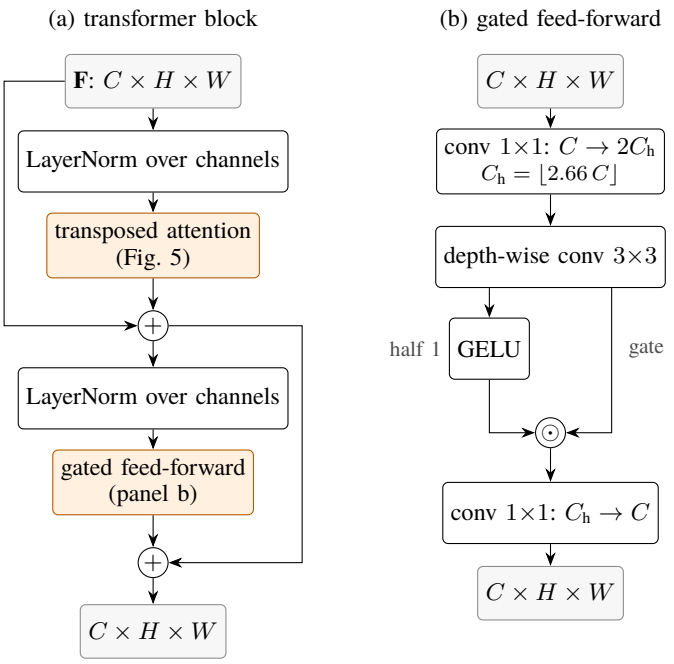


Fig. 4. (a) A transformer block of the prior: layer normalization over channels, transposed attention and a gated feed-forward network, each in a residual branch. (b) The gated feed-forward network: the expanded features are split in two halves, and the GELU of one half gates the other.

the two observations and that the proximal step depends on it. We therefore describe each input pair by a code $\mathbf{c}=(\sigma/\sigma_0-1,\ \delta/12,\ \log_{10}n_\mathrm{H}+2,\ \log_{10}n_\mathrm{M}+2)$, where (σ,δ) are the parameters of the operator used in the data steps and n is the noise standard deviation of an observation relative to its RMS, estimated from the observation itself as the median absolute diagonal second difference over all bands [67]. The step sizes become $\alpha_k=\mathrm{softplus}(a_k+\mathbf{u}_k^\top\mathbf{c})$ and $\beta_k=\mathrm{softplus}(b_k+\mathbf{v}_k^\top\mathbf{c})$, so that the two data terms can be weighted by their noise as in a maximum a posteriori estimate, and an embedding of $\mathbf{c}$ (two linear layers) scales and shifts the features of every prior (FiLM [68]) after its input convolution, at the input of its bottleneck and before its output convolution. All added parameters start at zero, so training starts from the unconditioned network; at the nominal operator without noise, $\mathbf{c}$ is nearly constant and the network reduces to PUFormer.

E. Self-ensemble

At test time we average the prediction for the input with the transposed prediction for the transposed input [69]. Transposition preserves the decimation phase, so both passes see observations generated by exactly the same operator. Flips and rotations do not: flipping a 256-pixel row maps sampled position $4m$ to $255-4m$, which has phase three, so we do not use them.

V. Experimental Setup

A. Data and protocol

Chikusei [70] is an airborne scene of 2517×2335 pixels with 128 bands from 363 to 1018 nm. The split sizes follow [15]: eight 256×256 test images, 64 validation patches and 832 training patches of 64×64. Because their locations are not specified there, we fix them as follows. We take the central 2048×2048 crop, which contains no no-data border, and divide it by its global maximum (15,133 digital numbers, DN). The test images are the eight 256×256 tiles of the top 256 rows, the validation patches tile rows 272–400, and training draws random 64×64 crops with dihedral augmentation from rows 416–2048 (800 non-overlapping patch equivalents). Sixteen-row gaps keep the three regions disjoint.

Pavia Centre [71] was recorded by the ROSIS sensor over the centre of Pavia, Italy; the distributed scene has 1096×715 pixels at 1.3 m and 102 bands between 430 and 860 nm. For this dataset, [15] uses the benchmark files released with PSRT [10]: 92 bands, a four-band multispectral image, four 256×256 test images and 142 validation and 1284 training patches of 64×64. These files are distributed through a file-sharing service that we could not access, so we rebuild the benchmark from the original scene to the same specification. We keep bands 11–102 (473–860 nm if the 102 bands are taken as evenly spaced over 430–860 nm), divide by the maximum of the retained bands (8,000 DN), apply the same blur and

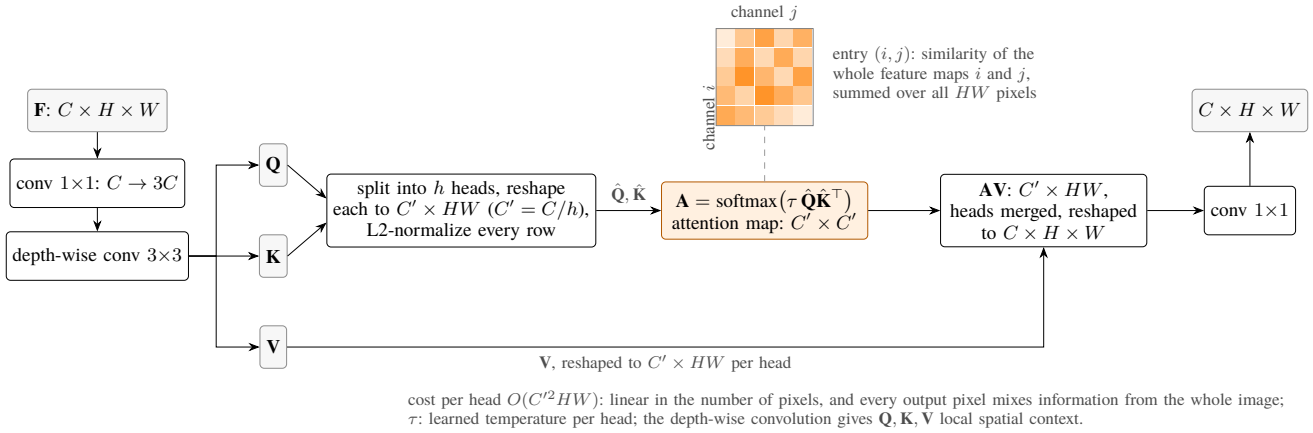


Fig. 5. Multi-Dconv-head transposed attention [44], the token mixer of the prior. Queries, keys and values are computed by point-wise and depth-wise convolutions, and the attention map is formed between the C' feature channels of each head rather than between the HW pixels, so its size does not depend on the image size. Replacing it by a depth-wise convolution changes the PSNR by less than 0.1 dB (Section VI-A).

decimation as for Chikusei, and simulate the HR-MSI with the IKONOS responses. The test images are the four 256×256 tiles of the top-left 512×512 region. The 63 validation patches lie on a 32-pixel grid in rows 528–656, and training draws random 64×64 crops with dihedral augmentation from rows 672–1096 and from the strip to the right of the test region (rows 0–512, columns 528–715), 97 non-overlapping patch equivalents in total. Gaps of 16 pixels keep the regions disjoint. As a check on the rebuilt data, our GSA implementation reaches 34.78 dB on our test images, against 35.35 dB reported for GSA in [15].

For both datasets we follow Wald's protocol [72]: the original image serves as ground truth and both observations are simulated from it, and every image, including every training crop, is degraded on its own with (1). Table I summarizes the two protocols.

Equal-budget benchmark. Reported numbers come from different splits, normalizations and metric code (Section V-B), so for the main comparison we retrain every learned method ourselves under one pipeline: SSR-NET [8], PSRT [10], DCT [54], MIMFormer [26] and DHIF-Net [20], built from their authors' code[1]. They cover a CNN, a pyramid window transformer, the dual-cross transformer that is among the strongest entries of [15], a multiscale mixer, and a model-guided unfolding network whose blur, decimation and spectral operators are learned convolution layers. Every method, PUFormer included, is trained for the same 4 h on one Tesla T4, with the same random patches and augmentation, batch size, EMA weights and validation-only model selection. The published networks keep their official optimizer settings (Adam, L1 loss, learning rate 10^{-4}, or 3×10^{-4} for DHIF-Net, no weight decay or gradient clipping) and are trained in 32-bit precision as released; DCT and MIMFormer use gradient checkpointing [73] to fit into the 16 GB of a T4, which does not change what they compute. The test metrics are computed by the same code for all methods.

[1]PSRT: github.com/Deng-shangqi/PSRT (commit 9c2be90); DCT: github.com/qingma2016/DCTransformer (bd75b15); MIMFormer: github.com/meiruni/MIMFormer (0904158); DHIF-Net: github.com/TaoHuang95/DHIF-Net (6434366). Only the numbers of input and output bands are changed; DHIF-Net's code covers ×8 and ×16, and at ×4 we use one of the two stride-4 layers of its ×16 operator.

TABLE I
THE TWO ×4 PROTOCOLS (DEGRADATION AND SPLIT SIZES OF [15]; IMAGE LOCATIONS OURS)

Item	Chikusei	Pavia Centre
HR-HSI	central 2048^2 crop, 128 bands	1096×715, bands 11–102 (92)
Normalization	÷ 15,133 DN	÷ 8,000 DN
LR-HSI	Gaussian 7×7, $\sigma = 2$; ×4 decimation; reflect border	
HR-MSI	WorldView-2, 8 bands	IKONOS, 4 bands
Test	8 of 256^2 (rows 0–256)	4 of 256^2 (rows, columns 0–512)
Validation	64 of 64^2 (rows 272–400)	63 of 64^2 (rows 528–656)
Training	random 64^2 crops, rows 416–2048	random 64^2 crops, see text
Selection	validation PSNR only	

B. Metrics

We report the eight metrics of [15], averaged over the test images: PSNR; SSIM [63]; the spectral angle mapper (SAM, degrees, averaged over the pixels whose ground-truth and estimated spectra are both nonzero) [74]; ERGAS with ratio 4 [75]; Q2n [76] on 32×32 blocks, computed on the image in DN exactly as in the reference implementation of the pan-sharpening toolbox [77], [78]; the band-averaged correlation coefficient (CC); the spatial correlation coefficient (SCC) [79] of Sobel gradient magnitudes as in the same toolbox; and RMSE in DN. The headline PSNR is $10\log_{10}(1/\mathrm{MSE})$ over the whole normalized image, i.e., its peak is the dataset maximum. The PSNR values of [15] also use a fixed peak: for every row of their Chikusei table, $\mathrm{RMSE}\cdot10^{\mathrm{PSNR}/20}$ is between 16.6k and 17.1k DN. Since our peak is lower (15,133 DN), our headline PSNR is conservative; with a 16.7k DN peak it would be 0.86 dB higher. For Pavia Centre the same product lies between 7.7k and 8.4k DN (median 8.07k), close to our peak of 8,000 DN, so there the PSNR values are on the same scale. The SSIM definition of [15] is not stated. We report SSIM as in the code released with PSRT [10], whose benchmark data [15] adopts (Gaussian 11×11 window, $\sigma = 1.5$, data range 1, i.e., the dataset peak), and additionally a stricter variant whose data range is each image's own maximum.

TABLE II
TRAINING RUNS

	Chikusei		Pavia Centre
	First run	v3 (final)	final
Initialization	random	first run	random
Border of $\mathcal{D}$	zero padding	reflect (exact)	reflect (exact)
Loss	L1 + 0.1 deep sup.	(17)	(17)
GPUs × batch	1×8	2×8	2×8
Peak learning rate	3×10^{-4}	1.5×10^{-4}	3×10^{-4}
Iterations	88,946	94,450	93,850
Training patches	0.71 M	1.51 M	1.50 M
Wall-clock time	9.5 h	10.9 h	10.2 h
Selected at	best val. (58.60 dB)	it. 44,000 (58.72 dB)	it. 43,000 (48.58 dB)

We also report observation consistency, $\mathrm{PSNR}(\mathcal{D}\hat{\mathbf{X}}, \mathbf{Y}_\mathrm{H})$ and $\mathrm{PSNR}(\mathcal{R}\hat{\mathbf{X}}, \mathbf{Y}_\mathrm{M})$.

C. Implementation

PUFormer is implemented in PyTorch [80] and trained on Kaggle NVIDIA Tesla T4 GPUs (Table II). On Chikusei we train in two runs. The first run trains from scratch on one GPU with zero-padded operators and the L1 losses only. The final model (v3) fine-tunes it with the exact reflect-border operators and the full loss (17) on two GPUs with distributed data parallelism [81]. On Pavia Centre, a single run trains the model from scratch with the exact operators and the full loss on two GPUs, within one 12-hour session that also runs all tests. All runs use AdamW [82] ($\beta = (0.9, 0.99)$, weight decay 10^{-4}), a 1000-iteration linear warm-up followed by a cosine decay [83] to 10^{-3} of the peak rate that ends at the time budget, and gradient-norm clipping at 0.1. The weights with the best validation PSNR are tested once. The ablation and benchmark runs (Sections VI-A and VI-B) train from scratch with these settings, the full loss and a single GPU for 4 h each; their models are evaluated without self-ensemble.

VI. RESULTS

A. Ablation study

Table III changes one module of the full model at a time and retrains each variant from scratch with the same budget (4 h on one T4, about 40,000 iterations for the full model). On Pavia Centre the full model and the zero-border variant were trained with three seeds; their PSNR varies by only 0.02 and 0.03 dB across seeds, which sets the scale for the other differences. Chikusei repeats every variant, again with three seeds for these two rows.

Data step and operator. Removing the data-consistency step together with the residual input (row (b)) costs 0.17 dB and lowers the agreement of the reconstruction with the LR-HSI from 77.6 to 68.9 dB. Keeping the exact kernel but replacing the reflect border by zero padding (row (d)) costs 0.08 dB on average over the three seeds, and each zero-border seed is below each full-model seed. A blur kernel and spectral response learned with the network (row (e)), the operator model of DHIF-Net and MHF-Net, cost 0.06 dB. Both variants reproduce the LR-HSI about 3 dB worse than the exact model (74.5 and 73.6 against 77.6 dB). This is what Theorem 3 predicts: their data steps move the ground truth, and the prior has to compensate. The compensation is tied to the operator used in training: Chikusei weights trained with a zero border fall from 58.60 to 49.52 dB validation PSNR when the exact border is substituted without retraining (Section S-I of the supplementary material). On the test images, the displacement of the ground truth by one LR-HSI data step, $\|\tilde{\mathcal{D}}^\top(\tilde{\mathcal{D}} - \mathcal{D})\mathbf{X}\|$ relative to $\|\tilde{\mathcal{D}}^\top\tilde{\mathcal{D}}\mathbf{X}\|$, is 4.66% for the zero border and 4.69% for the learned operator, and on 64×64 crops, where border pixels are a larger share, 11.11% and 11.17%. Chikusei (rows (o)–(z)) shows the same ordering with larger gaps: without the data step the PSNR is 0.30 dB lower, with a zero border 0.22 dB and with a learned operator 0.22 dB, and the agreement with the LR-HSI drops from 83.9 to 73.3, 75.9 and 77.1 dB.

The accuracy gap at the training blur is small; it becomes large when the blur changes (last column of Table III and Fig. 6). Given the true blur at test time, the exact model loses at most 0.13 dB for $\sigma \in \{1.5, 2.5, 3.0\}$, the zero-border model with the same swap 0.88 dB, and the learned operator, which cannot be replaced, 1.24 dB. Without the swap the exact model loses 7.50 dB: it is the most sensitive variant to a wrong operator, and the only one that fully recovers when given the right one. On Chikusei the difference is larger still: with the true blur the exact model loses at most 0.25 dB, the zero-border model 3.41 dB and the learned operator 3.45 dB. A zero border thus prevents the swap from working, even though it differs from the exact operator only at the image border, because the prior has learned to compensate a border error tied to the training blur.

Prior. Cross-stage memory (row (f)), the residual input (row (c)), the type of token mixer (rows (j) and (k): a depth-wise convolution [84] or self-attention within 8×8 windows [85] instead of transposed attention) and the number of stages (rows (g)–(i)) change the PSNR on Pavia Centre by at most 0.06 dB, comparable to the spread across seeds. A one-stage model with 3.33 M parameters is within 0.02 dB of the full model with 9.99 M; part of this parity comes from the fixed time budget, in which the smaller model completes almost three times as many iterations. The SAM and SSIM loss terms (row (l)) do not improve the PSNR, whereas on Pavia Centre deep supervision of the intermediate stages does (row (m); 0.12 dB lower without it). On Chikusei, where the three seeds of the full model differ by only 0.004 dB, the same modules change the PSNR by between -0.02 and +0.07 dB: memory +0.01, residual input +0.06, window attention +0.03, convolutional mixer +0.07, two and four stages -0.02 and +0.01, SAM and SSIM loss terms +0.02 and deep supervision +0.04 dB. Several of these simplifications are slightly better than the full model, so its prior is not optimal for this dataset, and deep supervision, which helps on Pavia Centre, does not help here. Only a single stage costs noticeably (0.20 dB): on the larger, 128-band problem at least two stages are needed. We keep the full model for all other experiments, but the ablation shows that PUFormer's accuracy depends more on its exact data-consistency steps than on the particular design of its transformer prior.

TABLE III

ABLATION STUDY: EACH ROW CHANGES ONE MODULE OF THE FULL MODEL (BOLD; ✓: USED, –: REMOVED) AND IS TRAINED FROM SCRATCH WITH THE SAME 4 H BUDGET ON ONE T4 (TEST SET, SINGLE PASS). Δ: PSNR CHANGE AGAINST THE FULL MODEL OF THE SAME DATASET. CONS.: PSNR($\mathcal{D}\hat{\mathbf{X}}$, $\mathbf{Y}_{\mathrm{H}}$)/PSNR($\mathcal{R}\hat{\mathbf{X}}$, $\mathbf{Y}_{\mathrm{M}}$) IN DB, WITH THE SIMULATION OPERATORS. SWAP: WORST TEST PSNR OVER BLUR $\sigma \in \{1.5, 2.5, 3.0\}$ WHEN THE TRUE BLUR IS GIVEN TO THE MODEL AT TEST TIME, MEASURED FOR THE OPERATOR VARIANTS; N/A: THE VARIANT HAS NO OPERATOR THAT COULD BE REPLACED. ROWS WITH ±: MEAN AND STANDARD DEVIATION OVER THREE SEEDS.

#	Data step	Res. input	Operator $\mathcal{D}, \mathcal{R}$	Memory	Mixer	K	SAM+SSIM	Deep sup.	Params (M)	PSNR	Δ	SAM	ERGAS	Cons. (dB)	Swap
Pavia Centre ×4															
(a)	✓	✓	exact	✓	MDTA	3	✓	✓	9.99	50.62±0.02	–	1.897	0.621	77.6/74.8	50.50
(b)	–	–	exact	✓	MDTA	3	✓	✓	9.99	50.46	-0.17	1.928	0.631	68.9/73.3	n/a
(c)	✓	–	exact	✓	MDTA	3	✓	✓	9.99	50.62	-0.01	1.900	0.621	75.7/73.8	–
(d)	✓	✓	**zero border**	✓	MDTA	3	✓	✓	9.99	50.54±0.03	-0.08	1.915	0.627	74.5/73.0	49.64
(e)	✓	✓	**learned**	✓	MDTA	3	✓	✓	9.99	50.57	-0.06	1.908	0.626	73.6/72.6	n/a
(f)	✓	✓	exact	–	MDTA	3	✓	✓	9.98	50.67	+0.05	1.898	0.618	77.2/74.8	–
(g)	✓	✓	exact	–	MDTA	**1**	✓	✓	3.33	50.60	-0.02	1.909	0.624	71.3/70.8	–
(h)	✓	✓	exact	✓	MDTA	**2**	✓	✓	6.66	50.64	+0.02	1.901	0.619	75.0/73.4	–
(i)	✓	✓	exact	✓	MDTA	**4**	✓	✓	13.32	50.68	+0.06	1.894	0.617	80.2/76.3	–
(j)	✓	✓	exact	✓	**conv**	3	✓	✓	9.94	50.60	-0.02	1.909	0.624	78.5/73.9	–
(k)	✓	✓	exact	✓	**window**	3	✓	✓	9.99	50.63	+0.01	1.898	0.622	78.6/75.5	–
(l)	✓	✓	exact	✓	MDTA	3	–	✓	9.99	50.66	+0.03	1.897	0.619	77.5/75.7	–
(m)	✓	✓	exact	✓	MDTA	3	✓	–	9.99	50.50	-0.12	1.900	0.629	77.7/74.6	–
Chikusei ×4															
(n)	✓	✓	exact	✓	MDTA	3	✓	✓	10.13	57.95±0.004	–	0.731	1.344	83.9/83.4	57.70
(o)	–	–	exact	✓	MDTA	3	✓	✓	10.13	57.65	-0.30	0.752	1.367	73.3/73.7	n/a
(p)	✓	–	exact	✓	MDTA	3	✓	✓	10.13	58.01	+0.06	0.727	1.337	82.8/82.8	–
(q)	✓	✓	**zero border**	✓	MDTA	3	✓	✓	10.13	57.73±0.05	-0.22	0.738	1.357	75.9/74.4	54.38
(r)	✓	✓	**learned**	✓	MDTA	3	✓	✓	10.13	57.73	-0.22	0.739	1.362	77.1/75.3	n/a
(s)	✓	✓	exact	–	MDTA	3	✓	✓	10.12	57.96	+0.01	0.728	1.342	84.1/83.0	–
(t)	✓	✓	exact	–	MDTA	**1**	✓	✓	3.37	57.75	-0.20	0.743	1.359	77.4/77.1	–
(u)	✓	✓	exact	✓	MDTA	**2**	✓	✓	6.75	57.92	-0.02	0.731	1.342	81.0/81.6	–
(v)	✓	✓	exact	✓	MDTA	**4**	✓	✓	13.51	57.96	+0.01	0.728	1.343	85.0/82.7	–
(w)	✓	✓	exact	✓	**conv**	3	✓	✓	10.09	58.02	+0.07	0.724	1.334	85.2/85.9	–
(x)	✓	✓	exact	✓	**window**	3	✓	✓	10.13	57.98	+0.03	0.725	1.339	84.4/85.0	–
(y)	✓	✓	exact	✓	MDTA	3	–	✓	10.13	57.97	+0.02	0.731	1.367	82.2/83.0	–
(z)	✓	✓	exact	✓	MDTA	3	✓	–	10.13	57.99	+0.04	0.727	1.336	84.1/83.7	–

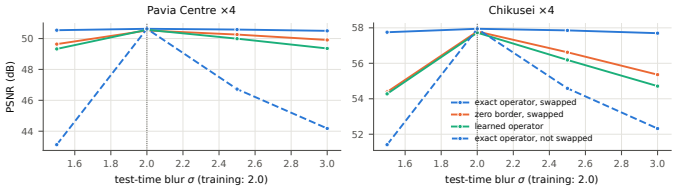


Fig. 6. Test PSNR when the blur of the test data differs from the training blur ($\sigma = 2$, dotted line), for the exact operator with and without the true blur supplied at test time, the zero-border operator with the true blur supplied, and the learned operator, which cannot be replaced.

B. Equal-budget benchmark

Table IV compares PUFormer with the five retrained networks; CC and SCC are listed in Table S-VI of the supplementary material. On Chikusei PUFormer reaches 57.95 dB, 0.99 dB above the best retrained network, and on Pavia Centre 50.63 dB (+1.00 dB), with 10% and 11% lower RMSE; it also has the lowest SAM and ERGAS on both datasets. MIMFormer is the strongest competitor on both. The unfolded DHIF-Net, whose operators are learned convolutions, reaches 54.39 and 47.67 dB; with 3,250 and 3,950 iterations in 4 h it is also the slowest network to train (Table IX).

Two checks address the fairness of a fixed budget. First, the retrained baselines reach or exceed the values reported for them in [15] although our data and splits differ: on Chikusei ×4, for example, MIMFormer reaches 56.96 dB against 53.36 dB reported and PSRT 53.99 dB against 50.67 dB, so our pipeline does not weaken them. Second, the budget favors fast networks: DCT completes only 4,800 iterations in 4 h on Pavia Centre, against 37,450 for PUFormer and 97,400 for MIMFormer. Doubling the budget to 8 h for PUFormer and the strongest competitor, MIMFormer, gives 50.67 and 49.97 dB on Pavia Centre ×4: the gap narrows from 1.00 to 0.69 dB but remains. The fixed budget also explains why the small SSR-NET, which completes its 300,000 iterations in about 1.5 h, ranks third on both datasets, ahead of the slower PSRT, DHIF-Net and DCT. On Chikusei ×4, PUFormer is more accurate than the best retrained network on all test images (one-sided Wilcoxon signed-rank test [86], $p = 0.0039$). Fig. **??** shows the error of every method on the test image of each dataset on which PUFormer is least accurate. GSA's error follows every edge; the errors of PSRT, DCT and DHIF-Net concentrate on the built-up areas and along building edges, MIMFormer's are smaller, and PUFormer's are the smallest and the most uniform.

C. Where the error lies

Theorem 2(c) splits the error of any fused image into a component in the observed subspace $\mathrm{row}(\mathcal{D}) + \mathrm{row}(\mathcal{R})$, which contradicts the observations, and a component in the fusion null space, which no data step can correct and only a prior can reduce. Table V applies this split to the test predictions of every method of the benchmark. Bicubic interpolation and GSA err mostly in the observed subspace: only 8.6% and 13.7% of their squared error lies in the null space on Chikusei, so they do not even reproduce their inputs. The learned

TABLE IV
EQUAL-BUDGET BENCHMARK: EVERY LEARNED METHOD RETRAINED BY US UNDER THE IDENTICAL PROTOCOL AND BUDGET (4 H ON ONE T4, THE SAME DATA, PATCHES, BATCH, EMA WEIGHTS AND VALIDATION-ONLY MODEL SELECTION; SINGLE PASS). GSA AND BICUBIC INTERPOLATION NEED NO TRAINING. BEST LEARNED METHOD IN BOLD; SSIM WITH THE DATA RANGE SET TO EACH IMAGE'S MAXIMUM; RMSE IN DN.

	Chikusei ×4						Pavia Centre ×4					
Method	PSNR↑	SSIM↑	SAM↓	ERGAS↓	Q2n↑	RMSE↓	PSNR↑	SSIM↑	SAM↓	ERGAS↓	Q2n↑	RMSE↓
Bicubic	33.63	0.7404	4.406	9.011	0.6914	321.81	25.52	0.6282	6.554	9.852	0.6806	436.32
GSA [4]	43.20	0.9778	2.730	3.279	0.9596	107.69	34.78	0.9438	5.229	3.494	0.9276	157.93
SSR-NET [8]	55.55	0.9958	0.954	1.707	0.9901	25.30	48.45	0.9954	2.443	0.758	0.9719	30.46
PSRT [10]	53.99	0.9948	1.057	1.794	0.9887	30.57	47.87	0.9958	2.212	0.801	0.9716	32.60
DCT [54]	53.18	0.9945	1.133	2.066	0.9871	33.58	46.94	0.9953	2.442	0.888	0.9712	36.26
MIMFormer [26]	56.96	0.9964	0.801	1.488	0.9918	21.65	49.63	0.9967	1.982	0.674	0.9738	26.66
DHIF-Net [20]	54.39	0.9953	1.020	1.660	0.9900	29.37	47.67	0.9956	2.295	0.840	0.9704	33.42
PUFormer	**57.95**	**0.9969**	**0.731**	**1.344**	**0.9929**	**19.45**	**50.63**	**0.9970**	**1.897**	**0.621**	**0.9754**	**23.68**

methods differ mainly in how much of the observed part they get wrong. PUFormer's error in the observed subspace is 8.6 times smaller than MIMFormer's on Chikusei and 5.8 times on Pavia Centre, and 98.5% of its remaining error lies in the null space. The split also attributes PUFormer's margin: correcting MIMFormer's observed part alone would raise its PSNR by 0.43 dB on Chikusei, and the remaining 0.62 dB come from a better estimate of the null-space component. The exact data steps account for the first part of the margin and the prior for the second; neither the attention mechanism nor any other single module of the prior is responsible for it (Section VI-A).

D. Comparison with reported results on Chikusei

The comparisons in this and the next subsection place PUFormer against the 17 methods tabulated in [15]. They are less controlled than Section VI-B: the other rows are copied from [15], our test images follow its protocol but are not necessarily the same pixels, and the PUFormer models here were trained longer (Table II). Table VI lists all Chikusei results of [15] together with PUFormer. The compared methods are GSA [4], CNMF [5], FUSE [7], U2Net [34], PSRT [10], DCT [54], DDIF [14], MIMFormer [26], MIMO-SST [11], OTPNet [35], SMGU [39], AMSF [27], SSDT [13], ASMNet [28], DPFormer [29], CLSNet [53] and the two-stage conditional diffusion model [15]; their values are copied from [15].

PUFormer ranks first of 18 on PSNR, SSIM (PSRT definition), SAM, ERGAS, CC, SCC and RMSE, with or without the self-ensemble. Relative to the best reported method, PSNR improves by 2.01 dB, RMSE by 26.4%, ERGAS by 12.2% and SAM by 1.1%. It ranks tenth on Q2n. Section S-III of the supplementary material shows that the Q2n deficit comes mainly from the eight ultraviolet bands (363–404 nm), which lie below the first WorldView-2 band and receive no high-resolution detail, and that the reported Q2n values do not follow PSNR the way our implementation does, so they were most likely computed differently. Because the RMSE is expressed in sensor digital numbers, it does not depend on any choice of peak, and it gives the most direct comparison across papers. Table S-IV of the supplementary material lists the long-trained models and the classical baselines under every PSNR and SSIM definition. Our GSA implementation scores 4.4 dB below the value reported in [15], so our protocol does not favor classical methods.

E. Comparison with reported results on Pavia Centre

Table VII compares PUFormer with the 17 methods of [15] on Pavia Centre. Trained from scratch in one session, PUFormer reaches 50.69 dB PSNR, 3.53 dB above the best reported value, and lowers the RMSE by 34.1% and ERGAS by 25.3%. It ranks first on six of the eight metrics (PSNR, SSIM in the PSRT definition, ERGAS, CC, SCC and RMSE), but 13th on SAM (1.883° against 1.4542°) and 16th on Q2n (0.9753 against 0.9987). With the stricter per-image data range its SSIM is 0.9970, below the reported 0.9972. The PSNR gain is larger than on Chikusei although the training region is eight times smaller (97 against 800 non-overlapping patch equivalents).

The Q2n shortfall on this dataset, and part of the SAM shortfall, come from one test image that contains a stretch of the river Ticino (63% water pixels; Section S-IV of the supplementary material). Water reflects little light, so its spectra are dominated by sensor noise that neither observation carries: over water PUFormer's SAM is 2.95°, over all other pixels 1.56°, and this image alone carries 97% of the Q2n deficit of the four test images. Whether the test images of [15] contain water is not stated, so on this dataset SAM and Q2n are the least comparable of the eight metrics and RMSE the most. As on Chikusei, our GSA implementation scores below the value reported for it (34.78 against 35.35 dB), so the rebuilt data do not favor our method.

F. Robustness of the long-trained models

The long-trained models of Tables VI and VII behave like the ablation models under a blur mismatch (Table S-II and Fig. S3 of the supplementary material). With the training operator, a blur with σ between 1.5 and 3.0 costs 3.9–7.5 dB on Chikusei and 3.3–6.8 dB on Pavia Centre; with the true

TABLE V

WHERE THE ERROR LIES (THEOREM 2). OBS. MSE: MEAN SQUARED ERROR IN THE OBSERVED SUBSPACE $\mathrm{row}(\mathcal{D}) + \mathrm{row}(\mathcal{R})$, THE PART THAT CONTRADICTS THE OBSERVATIONS ($\times 10^{-6}$, NORMALIZED INTENSITIES). NULL: SHARE OF THE SQUARED ERROR IN THE FUSION NULL SPACE, WHICH ONLY A PRIOR CAN REDUCE. $\mathrm{PSNR}_{\mathcal{N}}$: THE PSNR THE METHOD WOULD REACH IF ITS OBSERVED PART WERE EXACT. EQUAL-BUDGET MODELS, SINGLE PASS.

	Chikusei $\times 4$				Pavia Centre $\times 4$			
Method	PSNR	Obs. MSE	Null (%)	$\mathrm{PSNR}_{\mathcal{N}}$	PSNR	Obs. MSE	Null (%)	$\mathrm{PSNR}_{\mathcal{N}}$
Bicubic	33.63	431.001	8.6	44.27	25.52	2745.835	11.5	34.91
GSA	43.20	46.899	13.7	51.92	34.78	369.778	18.7	42.39
SSR-NET	55.55	0.703	74.9	56.83	48.45	2.049	85.8	49.11
PSRT	53.99	1.645	64.0	56.01	47.87	2.483	85.4	48.56
DCT	53.18	2.240	58.2	55.60	46.94	4.312	79.5	47.94
MIMFormer	56.95	0.186	90.6	57.38	49.63	0.430	96.3	49.79
DHIF-Net	54.39	1.546	66.1	56.32	47.67	3.980	78.1	48.75
PUFormer	57.94	0.022	98.5	58.01	50.63	0.074	99.2	50.66

TABLE VI

CHIKUSEI $\times 4$ COMPARISON. ALL ROWS EXCEPT PUFORMER ARE REPORTED IN [15]. BEST VALUE PER COLUMN IN BOLD. SSIM OF PUFORMER USES THE PSRT DEFINITION (DATA RANGE 1); WITH DATA RANGE EQUAL TO EACH IMAGE'S MAXIMUM IT IS 0.9969 (SINGLE PASS) AND 0.9970 (SELF-ENSEMBLE).

Method	PSNR↑	SSIM↑	SAM↓	ERGAS↓	Q2n↑	CC↑	SCC↑	RMSE (DN)↓
GSA [4]	47.5769	0.9933	1.7216	2.1667	0.9658	0.9894	0.9773	70.54
CNMF [5]	44.5291	0.9891	1.7259	2.8676	0.9662	0.9828	0.9645	101.64
FUSE [7]	41.1550	0.9787	2.5814	3.4710	0.9418	0.9736	0.9486	148.10
U2Net [34]	50.5455	0.9956	1.2389	1.8740	0.9928	0.9920	0.9798	49.97
PSRT [10]	50.6722	0.9952	1.2733	2.0107	0.9862	0.9906	0.9778	49.00
DCT [54]	52.1998	0.9977	0.9239	1.7448	0.9832	0.9935	0.9873	41.45
DDIF [14]	53.4421	0.9972	0.9387	1.8087	0.9963	0.9929	0.9840	35.58
MIMFormer [26]	53.3578	0.9973	0.9094	1.7611	0.9948	0.9931	0.9850	35.90
MIMO-SST [11]	53.6073	0.9972	0.9315	1.8119	0.9944	0.9926	0.9834	34.93
OTPNet [35]	52.8243	0.9969	1.0201	1.9332	0.9940	0.9916	0.9841	38.49
SMGU [39]	54.5244	0.9977	0.8719	1.6405	0.9952	0.9939	0.9875	31.29
AMSF [27]	49.3827	0.9944	1.4852	2.3943	0.9878	0.9879	0.9790	56.78
SSDT [13]	52.4788	0.9975	0.8768	1.7992	0.9939	0.9928	0.9847	40.00
ASMNet [28]	53.4528	0.9972	0.9472	1.9153	0.9874	0.9917	0.9848	35.89
DPFormer [29]	53.1632	0.9970	0.9483	1.8795	0.9949	0.9923	0.9824	37.18
CLSNet [53]	55.2007	0.9979	0.7836	1.6318	0.9943	0.9941	0.9869	29.06
Two-stage diffusion [15]	56.1871	0.9982	0.7203	1.5158	**0.9969**	0.9948	0.9891	25.73
PUFormer (single pass)	58.1467	**0.9989**	0.7171	1.3367	0.9930	**0.9967**	**0.9992**	19.06
PUFormer (self-ensemble)	**58.2016**	**0.9989**	**0.7126**	**1.3304**	0.9931	**0.9967**	**0.9992**	**18.94**

blur swapped in, it costs at most 0.26 and 0.11 dB. A spectral response shifted by 8 nm costs about 5 dB on Chikusei and is not compensated by swapping in the shifted response, because the priors rely on the training response beyond the data step; on Pavia Centre, whose IKONOS bands are 66–96 nm wide, the shift costs about 1 dB. Noise is the main weakness of models trained on noise-free data: at 30 dB SNR, PUFormer exceeds GSA by only 2.8 dB on Chikusei and by 6.0 dB on Pavia Centre.

G. Operators estimated from the test pair

The swaps in Sections VI-A and VI-F assume that the test blur is known. In practice it has to be estimated, as model-based methods [6] and blind deep fusion networks [87] do for the spatial and spectral responses. Because $\mathcal{D}$ acts on space and $\mathcal{R}$ on wavelength, they commute, and for the true blur $\mathcal{D}\mathbf{Y}_{\mathrm{M}} = \mathcal{R}\mathbf{Y}_{\mathrm{H}}$ holds exactly. We therefore estimate the blur width of each test image as $\hat{\sigma} = \arg\min_{\sigma} \|\mathcal{D}_{\sigma}\mathbf{Y}_{\mathrm{M}} - \mathcal{R}\mathbf{Y}_{\mathrm{H}}\|^2$ by a grid search with step 0.05, using only the two observations and the training spectral response, and give $\mathcal{D}_{\hat{\sigma}}$ to the network. Table VIII compares this with the training operator and the true one, without and with 35 dB noise on both inputs. On both datasets the estimate recovers the true width on every test image to within the grid step, also with noise, so the estimated and the true operator give the same reconstructions. In this simulated setting, where the blur is Gaussian and the spectral response known, the estimate is exact by construction; with a real sensor, a misspecified kernel shape or response would leave a residual error. With the estimate, the exact model's worst PSNR over the noise-free test blurs is 50.50 dB on Pavia

TABLE VII
PAVIA CENTRE ×4 COMPARISON. ALL ROWS EXCEPT PUFORMER ARE REPORTED IN [15] ON THE BENCHMARK DATA RELEASED WITH PSRT [10]; PUFORMER IS EVALUATED ON OUR SIMULATION OF THE SAME PROTOCOL FROM THE ORIGINAL SCENE (SECTION V). BEST VALUE PER COLUMN IN BOLD. SSIM OF PUFORMER USES THE PSRT DEFINITION (DATA RANGE 1); WITH DATA RANGE EQUAL TO EACH IMAGE'S MAXIMUM IT IS 0.9970 (SINGLE PASS) AND 0.9970 (SELF-ENSEMBLE).

Method	PSNR↑	SSIM↑	SAM↓	ERGAS↓	Q2n↑	CC↑	SCC↑	RMSE (DN)↓
GSA [4]	35.3495	0.9736	5.0812	2.7260	0.9818	0.9832	0.9768	137.78
CNMF [5]	32.1693	0.9469	3.8494	3.8446	0.9489	0.9683	0.9294	198.26
FUSE [7]	34.7418	0.9612	5.5628	3.2042	0.9103	0.9779	0.9469	146.90
U2Net [34]	44.3439	0.9963	1.6542	1.0924	0.9964	0.9973	0.9963	49.10
PSRT [10]	45.8875	0.9970	1.5230	0.9750	0.9974	0.9978	0.9966	41.85
DCT [54]	46.6745	0.9971	1.5169	0.8753	0.9973	0.9983	0.9972	37.66
DDIF [14]	46.2530	0.9970	1.5253	0.9125	0.9960	0.9982	0.9970	39.56
MIMFormer [26]	44.8497	0.9959	1.8225	1.0683	0.9957	0.9975	0.9954	46.19
MIMO-SST [11]	46.6334	0.9969	1.5587	0.8607	0.9975	0.9984	0.9972	37.68
OTPNet [35]	45.8081	0.9967	1.6573	0.9465	**0.9987**	0.9980	0.9968	41.51
SMGU [39]	47.1239	0.9972	1.4690	0.8547	0.9981	0.9984	0.9974	37.09
AMSF [27]	43.4465	0.9950	2.0533	1.2728	0.9929	0.9965	0.9940	54.25
SSDT [13]	46.3173	0.9965	1.6439	0.8918	0.9968	0.9982	0.9969	39.01
ASMNet [28]	46.2072	0.9966	1.6517	0.9019	0.9940	0.9982	0.9969	39.50
DPFormer [29]	42.4280	0.9948	1.9367	1.3986	0.9945	0.9957	0.9917	60.96
CLSNet [53]	46.5277	0.9969	1.5708	0.8412	0.9983	0.9983	0.9975	36.21
Two-stage diffusion [15]	47.1597	0.9972	**1.4542**	0.8262	0.9980	0.9985	0.9976	35.71
PUFormer (single pass)	50.6299	**0.9974**	1.8955	0.6206	0.9751	**0.9994**	**0.9995**	23.68
PUFormer (self-ensemble)	**50.6915**	**0.9974**	1.8833	**0.6170**	0.9753	**0.9994**	**0.9995**	**23.52**

Centre and 57.70 dB on Chikusei, against 43.13 and 51.41 dB with the training operator, 49.64 and 54.39 dB for the zero-border model with the same estimate, and 49.33 and 54.28 dB for the learned operator. Noise changes the picture. At 35 dB SNR, all variants fall to 44.4–44.8 dB on Pavia Centre and 48.7–49.5 dB on Chikusei, because none was trained with noise, and the operator models no longer differ: at the training blur, the exact model reaches 44.56 and 49.28 dB, the zero-border model 44.84 and 49.50 dB, and the learned one 44.77 and 49.37 dB. An exact data step enforces agreement with the observations, noise included, so its advantage requires training with noise when the inputs are noisy. The spectral response is not estimated: on Chikusei, swapping in even the true shifted response did not help (Section VI-F), because the priors depend on the training response beyond the data step.

H. Complexity

Table IX compares the parameters, operations, inference time and memory of all retrained networks on the same input. PUFormer has 10.13 M parameters and needs 296 GFLOPs for a 256×256 Chikusei image, which it reconstructs in 0.18 s on one Tesla T4 with mixed precision (92 bands: 0.18 s); the self-ensemble doubles this. It is slower than SSR-NET, MIMFormer and PSRT, but more than ten times faster than DCT and DHIF-Net. Because the attention is computed across channels, memory and time grow linearly with the number of pixels.

TABLE VIII
BLUR ESTIMATED FROM THE TEST PAIR ITSELF. $|\Delta\sigma|$: MEAN ERROR OF THE ESTIMATED BLUR WIDTH; PSNR (DB) WITH THE TRAINING OPERATOR (FIXED), THE ESTIMATED AND THE TRUE BLUR GIVEN TO THE MODEL, FOR THE EXACT OPERATOR, THE ZERO-BORDER MODEL (ESTIMATED) AND THE LEARNED OPERATOR. NOISE: 35 DB SNR ON BOTH INPUTS.

		Exact operator			Zero border	Learned
Test blur σ	$\lvert\Delta\sigma\rvert$	fixed	estimated	true	estimated	fixed
Pavia Centre ×4						
1.5	0.000	43.13	50.54	50.54	49.64	49.33
1.5 + noise	0.000	41.26	44.71	44.71	44.59	44.41
2.0	0.000	50.63	50.63	50.63	50.52	50.57
2.0 + noise	0.000	44.56	44.56	44.56	44.84	44.77
2.5	0.000	46.73	50.58	50.58	50.26	49.99
2.5 + noise	0.000	43.23	44.61	44.61	44.84	44.64
3.0	0.000	44.18	50.50	50.50	49.91	49.36
3.0 + noise	0.000	41.90	44.64	44.64	44.78	44.45
Chikusei ×4						
1.5	0.000	51.41	57.75	57.75	54.39	54.28
1.5 + noise	0.000	47.58	49.38	49.38	48.80	48.66
2.0	0.000	57.95	57.95	57.95	57.79	57.73
2.0 + noise	0.000	49.28	49.28	49.28	49.50	49.37
2.5	0.000	54.59	57.86	57.86	56.62	56.19
2.5 + noise	0.000	48.66	49.36	49.36	49.41	49.11
3.0	0.000	52.32	57.70	57.70	55.36	54.72
3.0 + noise	0.000	47.94	49.40	49.40	49.20	48.77

VII. DISCUSSION AND LIMITATIONS

What the exact operator buys. At the blur used for training, the exact operator improves PSNR only slightly over a zero-border or a learned operator, by 0.06 to 0.22 dB. Its clearer effects are a reconstruction that reproduces its inputs more closely, and a network whose operator can be exchanged at test time, which turns a blur mismatch from a loss of several decibels into one of a few tenths of a decibel at most, with the true or an estimated blur. The ablation also shows that the particular design of the transformer prior matters little: a convolutional mixer performs as well as transposed attention, cross-stage memory brings no measurable benefit, and a single stage costs at most 0.2 dB. We therefore see the exact operator as a property worth enforcing in any unfolded fusion network rather than as a feature of PUFormer's architecture.

Limitations. The study covers two airborne benchmarks at a scale factor of 4; other scenes, such as indoor images, and

TABLE IX
COMPLEXITY FOR ONE 256 × 256 CHIKUSEI IMAGE (128 BANDS, ×4) ON ONE TESLA T4.

Method	Params (M)	GFLOPs	Time (ms)	Memory (GB)
PUFormer	10.13	296	178	0.44
SSR-NET	0.44	58	11	0.11
PSRT	0.30	43	118	0.42
DCT	8.46	2640	2103	1.10
MIMFormer	16.46	224	62	0.45
DHIF-Net	48.67	14930	2938	2.08

larger scale factors remain to be tested. All experiments use simulated data, in which the operators are known up to the parameters we vary; on real sensor pairs the blur, the spectral response and the co-registration are only approximately known. The blur estimate of Section VI-G handles the first, but a mismatched spectral response is not compensated by swapping it, and training with randomized responses would be needed. The models are trained without noise, which costs accuracy at low signal-to-noise ratios (Section VI-F), and with noisy inputs the advantage of the exact operator disappears (Section VI-G); training with noise is needed before the operator's benefit can be expected on real data. The equal-budget benchmark gives every method the same 4 h on one GPU; this is fair in compute but favors fast networks, and although the longer-budget check and the comparison with reported values suggest that the ranking holds, longer training could narrow some gaps. The comparison with reported results in Tables VI and VII rests on test images that follow the published protocol but are not necessarily the same pixels, and on Pavia Centre data rebuilt from the original scene. There RMSE in digital numbers is the most robust basis, whereas SSIM with a per-image data range, Q2n and, on Pavia Centre, SAM are below the best reported values. Finally, the test sets are small (8 and 4 images), which limits per-image significance tests.

VIII. CONCLUSION

We studied how exactly the observation operators inside a deep unfolding network should match the operators that generated the data. With exact operators, border included, the ground truth is a fixed point of every data-consistency step; with a zero border or a learned operator it is not. In an ablation with repeated seeds, the exact operator gives a small but consistent accuracy gain, a reconstruction that agrees better with its inputs, and a network whose blur can be replaced at test time with almost no loss, also when the blur is estimated from the test images. Most components of the transformer prior, in contrast, can be removed or simplified at a cost of at most 0.2 dB. In an equal-budget benchmark against five published networks retrained from their authors' code, the resulting network, PUFormer, is the most accurate on Chikusei and Pavia Centre by 0.99 and 1.00 dB PSNR, and under the protocol of [15] it reaches 58.20 dB on Chikusei ×4 and 50.69 dB on Pavia Centre ×4. Applying exact operators to real sensor pairs with estimated blur and spectral response, and extending the comparison to other scenes and scale factors, are the main directions for future work.

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